Energy Conservation and Front Accuracy Across a Regularized Melting Interval

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Abstract

When a time step crosses phase endpoints, does an accurate linear solve also conserve the material's true energy? We connect a conservative enthalpy solver, an intentionally frozen apparent-capacity comparator, independent heat and Stefan references, and Python/C++ implementations in a reproducible verification study. On eleven fixed front settings, the conservative cumulative scaled energy defect stays below 7×10^{-13}. Both methods meet the prescribed 2% front-error target at the finest setting. Unfavorable comparisons and nonmonotone errors are retained: conservation does not imply uniformly superior front accuracy. Independent smooth heat tests separate first-order temporal and second-order spatial error. The contribution is an original connected verification study of established methods, not a new discretization or experimental physical validation.

1 Question, model and numerical connection

Enthalpy formulations for phase change are established [1]. Here we ask how replacing a finite enthalpy increment by an old-state tangent affects energy accounting and front accuracy when a step crosses a melting interval. DTU's Modelling in Materials and Manufacturing Engineering (MMME) topics on solidification and implicit conduction motivate this study. Equations, code and synthetic experiments are original; no private course files, exercise solutions or teaching code are distributed. The six imported Transport modules remain versioned snapshots. The new `phasechange` package and C++17 kernels form a connected study within that laboratory.

Let ρ, c, k, L denote constant density, specific heat, conductivity and latent heat per unit mass. On a stationary one-dimensional slab of area A, define volumetric enthalpy H and liquid fraction f:

$$H(T) = \rho c(T - T_{\mathrm{ref}}) + \rho L f(T), \qquad f(T) = \min\{1, \max[0, (T - T_s)/w]\}, \quad w = T_l - T_s > 0. \tag{1}$$

The model is $\partial_t H = k\partial_{xx} T + q$, with volumetric power q. There is no advection, density change, segregation or deformation. The constitutive inverse is exact on each branch. Clipping in f defines the material law; accepted temperatures are never clipped to hide errors and must be nonnegative kelvin.

For cell volume $V = A\Delta x$, internal conductance $g = kA/\Delta x$ and prescribed-wall conductance $2g$, conservative backward Euler solves

$$R_i = V[H(T_i^{n+1}) - H(T_i^n)] + \Delta t \sum_j g_{ij}(T_i^{n+1} - T_j^{n+1}) - \Delta t V q_i = 0. \tag{2}$$

The sum includes a prescribed wall as a fixed neighbor; insulation omits that face. Newton steps use a right-sided tangent and a positive definite tridiagonal Jacobian. A scalar line search zeros the directional derivative of the strictly convex residual potential when a full step passes its minimum. The stopping test is $\|R\|_\infty / [V(\rho c w + \rho L)] \le 10^{-10}$; iteration exhaustion is an error. The C++17 backend supplies residual, Jacobian and LDL^T solves through an explicit Python interface. Both backends share the Python safeguard and ledger; agreement checks implementation, not independent validity of that algorithm.

The frozen comparator replaces storage in (2) by $VH'_+(T_i^n)(T_i^{n+1} - T_i^n)$ for the entire step, including cooling from a phase endpoint. This deliberately uniterated approximation is not a claim about all apparent-capacity methods. Its linear residual can be small while the true enthalpy balance fails. With Q_j the signed boundary/source heat entering the slab in step j, define

$$D_n = \frac{|\sum_i V[H(T_i^n) - H(T_i^0)] - \sum_{j=1}^n Q_j|}{A\ell(\rho c w + \rho L)}, \qquad D_{\max} = \max_n D_n. \tag{3}$$

Reporting the largest transient defect avoids hiding drift through final-cycle cancellation. Algebraic closure and temperature/front accuracy are distinct.

2 Independent references and experimental protocol

Uniform heating and cooling remove spatial error. An independent event-traversal reference computes times to each phase endpoint and integrates with the appropriate constant capacity; it calls neither the enthalpy inverse nor the spatial solver. Six schedules with 1, 2, 4, 8 and 16 subdivisions give baseline temperature error scaled by the 2 K interval and energy error scaled by $\rho cw + \rho L$ of at most `0.000e+00` and `1.393e-14`. Constitutive inverse, monotonicity and endpoint tests complement this check.

For $L = 0$, an insulated slab has exact cell-average cosine amplitudes

$$a_i = \frac{\sin(\pi/(2N))}{\pi/(2N)} \cos(\pi x_i/\ell), \qquad \mu = \alpha(\pi/\ell)^2, \qquad \mu_h = \frac{4\alpha}{\Delta x^2} \sin^2 \frac{\pi}{2N}, \tag{4}$$

where $\alpha = k/(\rho c)$. Continuum and semidiscrete temperatures are $310 + 8a_i e^{-\mu t}$ and $310 + 8a_i e^{-\mu_h t}$ K. Backward Euler multiplies the discrete amplitude by $(1 + \mu_h \Delta t)^{-1}$ per step. Comparing numerical steps with exact semidiscrete evolution isolates temporal error; comparing the two exact exponentials isolates spatial error without time contamination. An independently assembled incidence matrix and dense matrix exponential check the discrete oracle; quadrature checks cell averages.

Synthetic constants are $\rho = 1200$ kg/m^3, $c = 900$ J/(kg K), $k = 1.8$ W/(m K), and $L = 90000$ J/kg when phase change is active. Heat studies use $\ell = 0.1$ m and $t = 1/\mu = 607.927101854$ s: 32 cells with 20–160 steps for time error, and 10–80 cells with exact discrete evolution for spatial error. These heat settings were specified on day 5; the Stefan protocol below was fixed on day 1. Table entries are maximum cell-temperature errors; successive order is $\log_2(E_{\text{coarse}}/E_{\text{fine}})$.

Steps	Time error (K)	Order	Cells	Space error (K)	Order
20	7.1907e-02	–	10	2.3829e-02	–
40	3.6322e-02	0.9853	20	6.0278e-03	1.9830
80	1.8255e-02	0.9926	40	1.5114e-03	1.9958
160	9.1510e-03	0.9963	80	3.7812e-04	1.9989

The largest error relative to the algebraic backward-Euler amplitude is `2.842e-13` K. These orders concern smooth constant-property heat conduction, not a moving phase front.

For a hot wall T_w and initially solid material at T_m, the independent one-phase sharp Stefan reference is

$$s(t) = 2\lambda\sqrt{\alpha t}, \qquad \lambda e^{\lambda^2} \operatorname{erf}(\lambda) = \frac{c(T_w - T_m)}{L\sqrt{\pi}}, \tag{5}$$

$$T(x,t) = T_w - (T_w - T_m)\frac{\operatorname{erf}(x/(2\sqrt{\alpha t}))}{\operatorname{erf}(\lambda)} \quad (x < s), \qquad T(x,t) = T_m \quad (x \geq s). \tag{6}$$

The constant-conductivity, prescribed-wall limit of the similarity construction is discussed in [2]; that work treats solidification and a broader boundary model. The melting sign convention here is checked independently through the heat PDE, boundary temperatures and Stefan flux jump. Bisection of a monotone logarithmic equation gives $\lambda = 0.3064239054$.

The fixed front protocol uses $T_m = 310$ K, $T_w = 330$ K, $A = 1$ m^2 and an insulated far wall at $\ell = 0.1$ m. Regularization has $T_s = T_m$, $T_l = T_m + w$ and zero initial liquid fraction. Observations at 30, 60, 90 and 120 s exclude the singular initial instant. The numerical front is the interpolated decreasing $f = 0.5$ crossing between cell centers; unresolved crossings are explicit failures. Four-level studies vary $N = 100, 200, 400, 800$, $\Delta t = 0.5, 0.25, 0.125, 0.0625$ s, or $w = 1, 0.5, 0.25, 0.125$ K, holding the other two at their finest values. Doubling ℓ and N preserves Δx. Shared finest cases are computed once: eleven settings, each with both methods and both backends.

3 Measured endpoint, energy and front behavior

Superscripts H, F label conservative enthalpy and frozen capacity. Uniform E_T is the largest error against the event reference. The interval is 309–311 K and the volume is 1 m^3. Solid heating starts at 305 K for 1 s; mushy heating/cooling start at 309.5/310.5 K for 2 s. Power is ± 2.7 MW/m^3. Crossing tests start at 305 or 330 K with the same signed power for 50 s. The closed cycle heats from 305 K for 50 s and cools for another 50 s. This table uses one step per segment; all subdivision histories remain in the JSON. Scientific notation uses `e` for powers of ten.

One step per segment	E_T^H (K)	E_T^F (K)	$D_{\max}^F$	D_{end}^F
solid heating	0.00e+00	0.00e+00	0.000e+00	0.000e+00
mushy heating	0.00e+00	0.00e+00	1.226e-14	1.226e-14
mushy cooling	0.00e+00	0.00e+00	1.226e-14	1.226e-14
cross both heating	0.00e+00	1.00e+02	9.804e-01	9.804e-01
cross both cooling	0.00e+00	1.00e+02	9.804e-01	9.804e-01
heat cool cycle	0.00e+00	1.00e+02	9.804e-01	0.000e+00

The 100 K error is missed latent heat divided by sensible capacity, L/c. Final-cycle cancellation hides transient drift. Within a constitutive branch the methods agree. These crossing failures concern the specified uniterated approximation and time steps, not all capacity formulations.

A connected slab test uses $\ell = 0.012$ m, $A = 0.003$ m^2, 24 cells, 305 K initial temperature and an insulated right wall. Five 10 s base steps use a 335 K left wall and $q = 2.7$ MW/m^3; five more use 295 K and negative power of the same magnitude. Each base step is subdivided uniformly. The temperature difference is between methods, not continuum error.

Subdivisions	$\max \lvert T_H - T_F \rvert$ (K)	$D_{\max}^H$	$D_{\max}^F$
1	69.453	2.451e-15	9.804e-01
2	73.183	4.171e-15	9.804e-01
4	74.972	1.399e-14	9.804e-01
8	68.604	9.861e-15	8.488e-01
16	34.821	1.961e-14	4.421e-01

The nonmonotone differences show why implementation agreement cannot determine accuracy. Ledgers retain all steps, including unsampled front-study steps. For that separate study, define

$$E_s = 100 \max_{t \in \{30,60,90,120\}} |s_h(t)/s(t) - 1|.$$

The shared finest case is labeled refined; numbered cases use the levels listed above. The doubled-domain case uses $\ell = 0.2$ m.

Setting	Cells	Δt (s)	w (K)	E_s^H (%)	E_s^F (%)
refined	800	0.0625	0.125	0.213862	1.542954
mesh-0	100	0.0625	0.125	2.861279	3.068805
mesh-1	200	0.0625	0.125	0.839380	1.032510
mesh-2	400	0.0625	0.125	0.625741	1.311259
time-0	800	0.5	0.125	0.217875	11.078534
time-1	800	0.25	0.125	0.214978	7.142458
time-2	800	0.125	0.125	0.214434	2.409037
width-0	800	0.0625	1	1.583802	2.223497
width-1	800	0.0625	0.5	0.898116	1.511194
width-2	800	0.0625	0.25	0.277012	1.551998
double-domain	1600	0.0625	0.125	0.213862	1.542954

Both refined methods meet the 2% target: $E_s^H = 0.213862\%$ and $E_s^F = 1.542954\%$. Across these settings the largest conservative and frozen $D_{\max}$ are `6.433e-13` and `4.248e-02`. The largest equation residual for either method is `2.977e-13` on its cell storage scale. A small linear residual does not imply true-energy closure; conservative closure does not remove discretization or regularization.

4 Interpretation, reproducibility and limits

Retained counterexamples. The finest conservative maximum front error is smaller, but this is not a universal ranking. At 30 s with 200 cells, frozen error is 0.177702% versus 0.699333% for enthalpy. On 800 cells with $w = 0.125$ K, reducing Δt from 0.5 to 0.0625 s increases early conservative error from 0.030316% to 0.066729%. This is consistent with error cancellation, not evidence that coarse steps are generally preferable. No Stefan convergence order is asserted. Shrinking w at fixed resolution can eventually underresolve the phase interval; the width study is a sensitivity experiment, not a complete asymptotic limit.

Independent evidence and its limits. Event tests verify local accounting. Python/C++ residual/state comparisons, finite-difference derivatives away from kinks, one-sided endpoint checks and native invariants assess implementation consistency. Exact discrete heat evolution separates temporal from smooth spatial continuum error. Sharp Stefan discrepancies include finite-interval regularization and front interpolation as well as discretization. These factors are varied separately; an additive error decomposition is not claimed. Doubling the domain produces zero measured front and common-cell temperature differences for both methods at the four observations. This supports insensitivity at floating-point resolution in this window only, not a bound for later times or other inputs. Residuals are below the stopping threshold; no tolerance sweep was performed.

Reproduction and evidence chain. Source and synthetic results are public [3]. Use Python 3.11 or later (CI: 3.12), NumPy/SciPy and a C++17 compiler. In a fresh virtual environment, run `python suite.py` with `install`, `verify`, `native` and `test` in turn. Reproduce the baseline with `python -m phasechange`, the comparison with `python -m phasechange --study comparison`, and refinement with `python -m phasechange.refinement`. The comparison and refinement commands accept `--backend native` and `--check-against` with a saved JSON path. Use distinct `--output` paths to preserve reference data. Linux and Windows CI repeat both complete campaigns, storing sampled fields, signed heat histories, front status, acceptance flags and numerical source/protocol SHA-256 hashes. Tests fail on missing evidence or numerical disagreement. The six imported snapshots remain separately installable and hash-checked; their prior synthesis remains at the v0.1.0 tag. Original histories are preserved.

After reproducing changed experiments, run `python manuscript.py generate`. It refuses stale numerical source hashes and generates all five table/number fragments. `python manuscript.py check` rejects stale tables. Build with `python suite.py paper --engine tectonic` or pdfLaTeX; the build rejects unresolved references, overfull boxes and more than five pages including references. `python manuscript.py check-report` checks page count and the complete source/data/table/PDF hash set. Mutation tests cover source drift, altered measurements, missing observations, failed gates, changed PDFs and incomplete manifests. CI verifies the committed artifact before rebuilding it independently. Local and CI pages are rendered for visual review, with actual outcomes in dated validation records.

Conclusion and scope. Finite enthalpy increments preserve energy across phase endpoints here; conservation cannot replace independent accuracy checks. The model excludes convection, segregation, unequal phase properties and hysteresis. No calibration, experimental validation, speedup or sanitizer coverage is claimed. Release remains conditional on all acceptance gates.

References

[1] V. R. Voller, M. Cross and N. C. Markatos. An enthalpy method for convection/diffusion phase change. *International Journal for Numerical Methods in Engineering* 24(1), 271–284 (1987). https://doi.org/10.1002/nme.1620240119.

[2] A. N. Ceretani, N. N. Salva and D. A. Tarzia. An exact solution to a Stefan problem with variable thermal conductivity and a Robin boundary condition (2017). https://arxiv.org/abs/1706.06984.

[3] Transport Verification Lab. Versioned source, synthetic experiments and validation records. https://github.com/racostacme-come/transport-verification-lab. The manuscript build report pins source and result SHA-256 hashes.